EXPERIMENT 25

Recognize a watch in an image using basic object recognition with OpenCV.

Aim

Recognize a watch in an image using basic object recognition with OpenCV.

Algorithm

• Import OpenCV and load the scene and watch template.

• Convert both images to grayscale.

• Match the template against the scene.

• Find the best matching location and compare its score with a threshold.

• Draw a rectangle around the detected watch and save the result.

Code

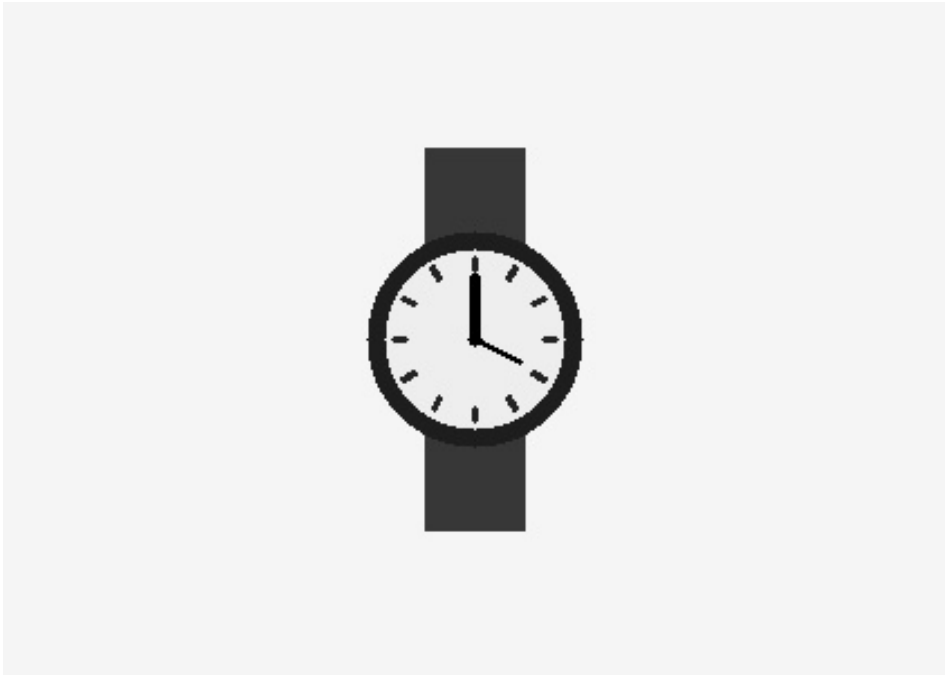
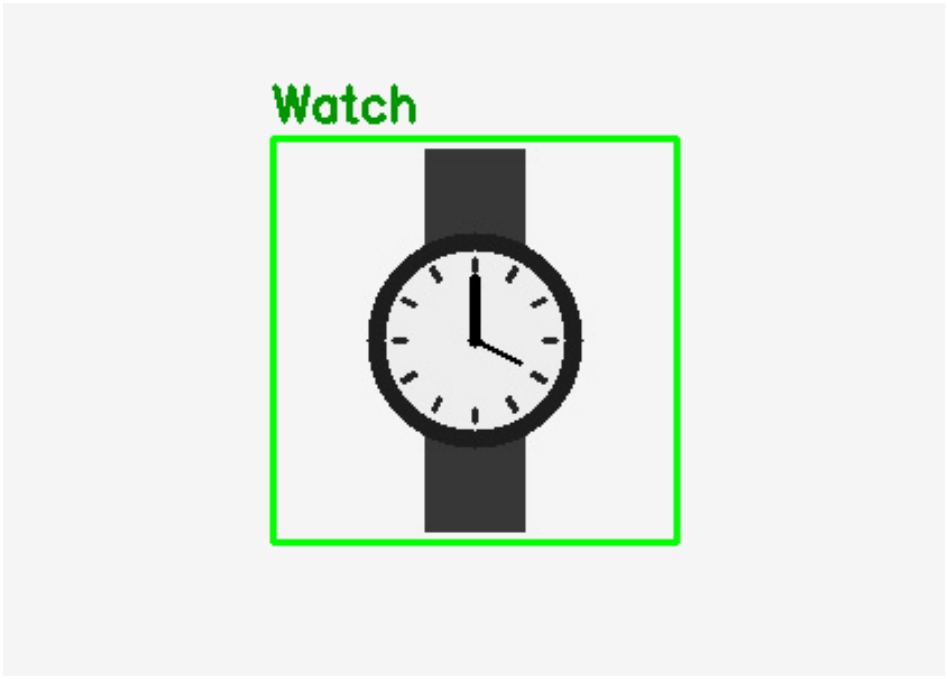
```
import cv2

img = cv2.imread("images/watch_scene.jpg")
template = cv2.imread("images/watch_template.jpg")
if img is None or template is None:
    raise FileNotFoundError("Watch images not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
temp = cv2.cvtColor(template, cv2.COLOR_BGR2GRAY)
h, w = temp.shape
match = cv2.matchTemplate(gray, temp, cv2.TM_CCOEFF_NORMED)
_, score, _, location = cv2.minMaxLoc(match)
result = img.copy()
if score &gt; 0.5:
    x, y = location
    cv2.rectangle(result, (x, y), (x + w, y + h), (0, 255, 0), 2)
    cv2.putText(result, "Watch", (x, max(20, y - 8)), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 150, 0), 2)
cv2.imwrite("output_images/experiment25_watch_recognition.png", result)
print(f"Match score: {score:.2f}")
print("Saved output_images/experiment25_watch_recognition.png")
```

Input Image:

Output Image:

Note: This is template matching, which recognizes a watch similar to the supplied template; it is not a trained general-purpose object detector.

Input Image



Output Image